

Thesis Proposal

Topic: Web-Based automated Comic Generation from Text (Single Character Focus)

Project Idea:

The project aims to create a web-based system that generates short comic book sequences from user-provided text descriptions, focusing on a single character. The main goal is to simplify the comic creation process by limiting the output to a few pages and maintaining visual consistency for one character.

How It Works:

The system will utilize two pre-trained AI models:

1. **Image Generation Model (Stable Diffusion or Similar):** This model will convert the user's text input into comic-style images. Each frame or panel will be generated to fit the visual style of a comic book.
2. **Text Prompt Processing Model:** This second model will handle the text input provided by the user and structure it in a way that ensures the generated images maintain consistency across the comic panels, resembling the flow and style of a comic book. However, the comic book will contain **images only**, with no written text in the panels.
3. **Flaw:** To reduce the complexity one story can be generated in many ways and it is possible they won't be able to find the last story (previous comic pages) if they don't save it.

The AI models will run on a local machine (my personal computer), and I will handle any commercial services if required for additional computational resources.

Front-End (Website Features):

The system will be accessible via a **user-friendly website**, which will offer the following features:

- **Text Input Interface:** Users will be able to input their story in a text box to submit in a form style.
- **Image Preview:** Once the comic pages are generated, users can save them for later use.
- **Downloadable PDF:** After previewing, users can download their comic as a **PDF file** for personal use or sharing.
- **Email Delivery:** Users will have the option to receive their generated comic as a PDF via **email**.
- **User Registration and Login System:** User can create account and later login or logout from it.
- **User History:** A **database** will keep track of each user's generated comics, allowing them to revisit, redownload, or delete past projects.
- **Payment System (Optional Feature):** For premium users or those who want more advanced features (e.g., higher-quality images, more pages), a **payment system** could be implemented.

Conclusion:

- This project is designed as a **proof-of-concept** to demonstrate how AI can assist in generating comic-style images from text. The focus on a single character and limited page output makes the project manageable while allowing users to create visually consistent comics. Although the system will not generate complete comic books with text bubbles or dialogue, it lays the groundwork for future developments in automated comic creation using AI.